Reference: the 8 % zero-point bias in the fdamador neutrino run

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A companion to the report on the fdamador neutrino, recording one error found in Part 4 and how it was handled, so others can judge the fix for themselves.

Summary

An 8 % excess in the zero-point run was a simulation artifact, now removed and shown not to depend on the fix.

- **What went wrong:** each run started the noise at a random voltage, and the start-up operating point froze that voltage onto the ring's capacitors. The frozen energy, on average $\hbar\omega_0/(\pi Q)$, was read as part of the zero-point energy.
- **How it was found:** the measured stored energy sat above the expected value computed beside it, by 2.9σ at Q = 10.2. The energy ledger itself stayed balanced throughout.
- **The fix:** the noise is switched on with a smooth ramp from zero, inside the settling time that is already discarded.
- **The check:** on identical noise, ramps of 2Q, 5Q and 8Q agree to four digits and match the expected value within 0.6σ, while an abrupt start reproduces the bias exactly.

The ramp is a simulation technique, not physics from neutrino papers, and it was added after the mismatch was seen. Both points are stated plainly below so others can judge.

Record of events

All times are Hawaii time on 4 November 2026, newest first. The run is Part 4 of the fdamador neutrino report: a scaled ring ($\omega_0 = 1$, $\hbar = 1$) driven by its radiation resistance's zero-point noise.

Time	Step	What happened
23:06	Independence check	Ramp lengths of 0, 2Q, 5Q and 8Q run on identical noise (results below)

Time	Step	What happened
22:58	Question raised	Asked whether the ramp was taken from nature or added to make results agree (answer below)
22:54	Wording corrected	The report said the ledger caught the error; it was the expected value. Changed in the report
22:43	Fix verified	With the ramp, 12 runs each: Q = 10.2 gave 0.4815 ± 0.0005 against 0.4817; Q = 40.7 gave 0.4900 ± 0.0031 against 0.4916
22:42	Cause found	Quarter-by-quarter energies showed a fixed offset; traced to a frozen DC charge from the start-up operating point
22:41	Bias confirmed	12 runs each: Q = 10.2 read 0.5230 ± 0.0144 (2.9σ high), Q = 40.7 read 0.5004 ± 0.0051 (1.7σ high)
22:40	Step size ruled out	A quarter of the time step gave an identical 0.5287, so time-stepping was not the cause
22:39	Seed spread seen	Three seeds at Q = 10.2 gave 0.483, 0.529 and 0.563, wider than their own error bars
22:38	First anomaly	Q = 10.2 read 0.5203 ± 0.0009 against 0.4817 expected, 8 % high, with only 2 runs

The bias as equations

The excess was a DC charge frozen onto the ring's capacitors at the start of each run, worth on average $\hbar\omega_0/(\pi Q)$. It was real energy in the wrong place, not energy from nowhere.

1. Where it came from. The noise voltage starts at a random value $v(0) = v_0$. kronspice begins a transient from the DC operating point, where inductors are shorts and capacitors open circuits, so the series capacitors take the whole source voltage:

$$\sum_k v_k = v_0 \quad\Rightarrow\quad E_{off} = \tfrac{1}{2} C v_0^2 \qquad (C = 1\ \text{F, the ring's total})$$

2. Why it stayed. The capacitors can discharge only through the 1 GΩ DC bleeders, with a time constant far longer than the run:

$$\tau = R_B C_k = 10^9\ \Omega \times 4\ \text{F} = 4 \times 10^9\ \text{s} \gg 8{,}192\ \text{s}$$

So each stored-energy reading was E_zp + E_off.

3. How big it was. v_0 is one sample of the noise, so its mean square is the noise power summed over the band, with one-sided spectrum $S_V = \hbar\omega R/\pi$, band edge $\Omega = 2\omega_0$ and $R = 1/Q$:

$$\langle v_0^2 \rangle = \int_0^{\Omega} \frac{\hbar \omega R}{\pi}\, d\omega = \frac{\hbar R \Omega^2}{2\pi} = \frac{2\hbar\omega_0}{\pi Q} \quad\Rightarrow\quad \langle E_{off} \rangle = \frac{\hbar \omega_0}{\pi Q}$$

Q	Predicted offset ($\hbar\omega_0$)	Share of ½$\hbar\omega_0$	Observed excess before the fix ($\hbar\omega_0$)
10.2	0.031	6.5 %	+0.041 (12 runs)
40.7	0.0078	1.6 %	+0.009 (12 runs)
203	0.0016	0.3 %	not resolved (scatter ±0.03)

Each run's offset comes from one random draw of v_0^2, which is often near zero and sometimes several times its mean. That explains why three seeds at Q = 10.2 gave 0.483, 0.529 and 0.563.

4. Why the ledger still balanced. The source paid for the offset at t = 0, and it was booked as energy stored at the start:

$$E_{zp,in} = E_{rad,out} + (E_{end} - E_{start}), \qquad E_{start} = E_{off}$$

The ledger had no residual. What exposed the error was comparing the measured stored energy with the expected value computed beside it.

The fix, and why it is not a fit

The noise is multiplied by a smooth ramp that starts at exactly zero, so the start-up operating point charges nothing and E_off = 0.

$$v_{used}(t) = v(t)\, w(t), \qquad w(t) = \begin{cases} \tfrac{1}{2}\left(1 - \cos\dfrac{\pi t}{T_r}\right) & t < T_r \\ 1 & t \ge T_r \end{cases}, \qquad T_r = 5Q$$

In the code (`zpf.go`), applied to each noise realization before its circuit is built:

```
ramp := z.Ramp * z.Q // z.Ramp = 5 by default
for i := range v {
    if t := float64(i) * z.Dt; t < ramp {
        v[i] *= 0.5 * (1 - math.Cos(math.Pi*t/ramp))
    }
}
```

- **Starting at zero** removes the frozen charge entirely, rather than shrinking it.
- **A raised cosine** has zero slope at both ends, so the switch-on adds no kick of its own. An abrupt switch-on would be a step and leave its own charge.
- **Five Q** is long enough for the ring to follow gently. It ends inside the first 10Q of every run, which was already discarded as settling time before averaging, so the averaged window and the expected value are unchanged.

Is the ramp found in nature or in neutrino papers? No. It is a simulation technique, and it corrects an artifact of simulating rather than adding physics. A real zero-point field has no switch-on moment: it has always been there. A simulation must start at some t = 0, and starting abruptly froze a DC charge the real, always-on field would never produce, so ramping up from zero is the closer imitation of "always on". The same idea appears as adiabatic switching of interactions in quantum field theory and as windowing in signal processing.

Why it is not a fit to the answer:

- The bias was predicted quantitatively, as $\hbar\omega_0/(\pi Q)$, before the fix, matching both its trend with Q and the scatter between seeds.
- No parameter was adjusted toward the target. The ramp length was set once, to 5Q, so that it ends inside the discarded window.
- One fix removes the bias at every Q, from 6.5 % at Q = 10.2 to 0.3 % at Q = 203.

What an evaluator should weigh: the ramp was added after the mismatch was seen. The independence check below tests whether it steers the result.

Independence check: does the ramp steer the result?

No. Any ramp from 2Q to 8Q gives the same stored energy to four digits, and starting without one reproduces the original bias exactly.

Each row is 12 kronspice realizations on identical noise (seed 1951), so only the ramp differs between rows of the same Q. Stored energy is in units of $\hbar\omega_0$; deviation is (measured − expected) / standard error.

Q	Ramp	Stored	Expected	Deviation	Zero-point in / radiated out
10.2	none (abrupt start)	0.5230 ± 0.0144	0.4817	+2.9σ	1.0011
10.2	2Q	0.4815 ± 0.0005	0.4817	−0.4σ	1.0012
10.2	5Q (used)	0.4815 ± 0.0005	0.4817	−0.4σ	1.0012
10.2	8Q	0.4814 ± 0.0005	0.4817	−0.4σ	1.0012
40.7	none (abrupt start)	0.5004 ± 0.0051	0.4916	+1.7σ	1.0051
40.7	2Q	0.4900 ± 0.0031	0.4916	−0.5σ	1.0052
40.7	5Q (used)	0.4900 ± 0.0031	0.4916	−0.5σ	1.0053
40.7	8Q	0.4899 ± 0.0031	0.4916	−0.6σ	1.0053

What this shows:

- **The bias is the start-up charge, not the ramp's shape.** Every ramp length removes it equally; the abrupt start brings back exactly the 0.5230 and 0.5004 first measured.
- **The ramp length is not a tuning knob.** Quadrupling it, from 2Q to 8Q, moves the result by 0.0001 $\hbar\omega_0$, far inside the error bars.
- **The inflow/outflow ratio barely moves** across all rows, because the frozen charge was booked as stored energy, not as flow.

This check is now a permanent test, `TestRampIndependence`, which runs ramps of 0, 2Q and 8Q on the same noise and fails if the ramped results differ or if the abrupt start stops showing the bias.

Limits and open points

These hold whatever the ramp check shows, and an evaluator should weigh them alongside the results.

- **Error bars from few runs are unreliable.** With 2 or 3 realizations the quoted standard error was too small: one early point read ± 0.0009 while seeds disagreed by 0.08. Quoted results use 12 runs for this reason, and the per-run spread is still uneven.

- **½ħω₀ is approached, not hit.** The expected value is 0.4817–0.4942 rather than 0.5, because the noise stops at $2\omega_0$ and the waveform is joined by straight lines between samples (a sinc^4 loss). Each result is compared with the expected value for the same noise modes, not with 0.5.
- **Time-stepping error is about 7×10^{-4}** of the radiated energy. It sits in the ledger's mismatch term and is small against every effect discussed here.
- **Units are scaled.** The ring runs with $\omega_0 = 1$ rad/s and $\hbar = 1$. Because the circuit is linear, energies scale with ħ, but this is a statement about the model's mathematics, not a measurement at the neutrino's 0.1 eV.
- **An earlier claim was wrong and corrected.** The report first said the ledger caught the error; the ledger stayed balanced, and the expected value exposed it. The report was edited on 4 November 2026 at 22:54.
- **The 1 GΩ bleeders are a model choice.** They give the ring a DC path to ground. They are why a start-up charge freezes, and they are a million times larger than the ring's impedance, so they do not affect the oscillation itself.

How to reproduce

Everything here runs from `examples/fdamador_neutrino` in fdamador/kronspice, branch `fdamador-neutrino`, commit ff182cd (local; not yet pushed). Runs are deterministic: the same seed gives the same numbers.

What	Where
Zero-point model and the ramp	`zpf.go`: `ZPFRing`, field `Ramp` (0 = abrupt start, default 5)
Independence test	`TestRampIndependence` in `main_test.go`
Inflow test	`TestZeroPointInflow` in `main_test.go`
Raw rows of the check	`ramp_check.txt` (seed 1951, 12 realizations per row)
Report this document accompanies	The fdamador neutrino: a report

```
go test ./examples/fdamador_neutrino -run
'TestRampIndependence|TestZeroPointInflow' -v
go run ./examples/fdamador_neutrino   # Part 4 prints the zero-point results
```

To see the bias yourself, set `Ramp: 0` in `DefaultZPFRing` and run Part 4: the Q = 10.2 point rises back above its expected value.